

Telecom Customer Churn Analysis

End-to-End Analytics Project Stage 2: SQL — Deep EDA + Analysis

1. Stage 2 Overview

In Stage 2, the cleaned dataset from Excel was imported into MySQL for deep exploratory data analysis. A total of 13 SQL queries were written to uncover churn patterns across multiple dimensions including contract type, internet type, revenue impact, churn reasons, customer demographics, add-on services, referrals and geographic distribution.

Tools & Environment

Database: MySQL 8.0 | Client: MySQL Workbench | Dataset: 7,043 rows, 41 columns | Import method: LOAD DATA INFILE with utf8mb4 character set

2. Database Setup

Create Database & Import

```
CREATE DATABASE telecom_churn;
USE telecom_churn;

-- Table created with 41 columns matching cleaned Excel file
-- Imported using LOAD DATA INFILE with CHARACTER SET utf8mb4
-- to handle BOM character at start of CSV file

LOAD DATA INFILE 'path/to/Telecom_churn_clean.csv'
INTO TABLE telecom_churn
CHARACTER SET utf8mb4
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\r\n'
IGNORE 1 LINES;
```

Verification

```
SELECT COUNT(*) FROM telecom_churn;
-- Result: 7,043 rows confirmed
```

3. SQL Queries & Findings

Query 1 — Overall Churn Rate

The first query establishes the baseline — what percentage of customers churned, stayed or joined.

```

SELECT
    customer_status,
    COUNT(customer_id) AS total_customers,
    ROUND(COUNT(customer_id) * 100.0 / SUM(COUNT(customer_id)) OVER(), 2) AS
percentage
FROM telecom_churn
GROUP BY customer_status
ORDER BY percentage DESC;

```

Customer Status	Total Customers	Percentage
Stayed	4,720	67.02%
Churned	1,869	26.54%
Joined	454	6.45%

Insight: 26.54% churn rate — above industry average

Over 1 in 4 customers churned. The company is losing customers nearly 4x faster than it gains new ones (26.54% churned vs 6.45% joined) — a dangerous long-term trend.

Recommendation: Treat churn reduction as a top business priority. Focus retention efforts on the 1,869 churned customer segment to recover lost revenue.

Query 2 — Churn Rate by Contract Type

Analyses how contract type influences customer churn using a CASE WHEN conditional counter.

```

SELECT
    contract,
    COUNT(customer_id) AS total_customers,
    SUM(CASE WHEN customer_status = 'Churned' THEN 1 ELSE 0 END) AS
churned_customers,
    ROUND(SUM(CASE WHEN customer_status = 'Churned' THEN 1 ELSE 0 END) * 100.0 /
COUNT(customer_id), 2) AS churn_rate
FROM telecom_churn
GROUP BY contract
ORDER BY churn_rate DESC;

```

Contract Type	Total Customers	Churned	Churn Rate
Month-to-Month	3,610	1,655	45.84%
One Year	1,550	166	10.71%
Two Year	1,883	48	2.55%

Insight: Contract type is the strongest churn predictor

Month-to-Month customers churn at 45.84% — 18x higher than Two Year contract customers at 2.55%. The longer the commitment, the stronger the retention.

Recommendation: Aggressively incentivize Month-to-Month customers to upgrade to longer contracts through targeted discounts and loyalty rewards.

Query 3 — Churn Rate by Internet Type

Examines churn patterns across different internet service types.

```
SELECT
  internet_type,
  COUNT(customer_id) AS total_customers,
  SUM(CASE WHEN customer_status = 'Churned' THEN 1 ELSE 0 END) AS
  churned_customers,
  ROUND(SUM(CASE WHEN customer_status = 'Churned' THEN 1 ELSE 0 END) * 100.0 /
  COUNT(customer_id), 2) AS churn_rate
FROM telecom_churn
GROUP BY internet_type
ORDER BY churn_rate DESC;
```

Internet Type	Total Customers	Churned	Churn Rate
Fiber Optic	3,035	1,236	40.72%
Cable	830	213	25.66%
DSL	1,652	307	18.58%
No Internet Service	1,526	113	7.40%

Insight: Fiber Optic — the premium service — has the highest churn

40.72% of Fiber Optic customers churned despite paying premium prices. Customers are finding better value with competitors — confirmed by churn reason analysis.

Recommendation: Urgently review Fiber Optic pricing and service quality. Deploy targeted retention campaigns for Fiber Optic Month-to-Month customers.

Query 4 — Revenue Lost to Churn

Quantifies the financial impact of churn across all customer status groups.

```
SELECT
  customer_status,
  COUNT(customer_id) AS total_customers,
  ROUND(SUM(total_revenue), 2) AS total_revenue,
  ROUND(AVG(total_revenue), 2) AS avg_revenue_per_customer,
  ROUND(SUM(monthly_charge), 2) AS total_monthly_charges,
  ROUND(AVG(monthly_charge), 2) AS avg_monthly_charge
FROM telecom_churn
GROUP BY customer_status
ORDER BY total_revenue DESC;
```

Status	Total Customers	Total Revenue	Avg Revenue	Total Monthly	Avg Monthly
Stayed	4,720	\$17,632,392.12	\$3,735.68	\$291,400.60	\$61.74
Churned	1,869	\$3,684,459.82	\$1,971.35	\$137,086.65	\$73.35
Joined	454	\$54,279.75	\$119.56	\$19,420.30	\$42.78

Insight: Churned customers paid the most but generated the least total revenue

The company lost \$3,684,459.82 in total revenue and \$137,086.65 in monthly recurring revenue to churn. Churned customers paid \$73.35/month on average — \$11.61 more than stayed customers — yet generated 47% less total revenue because they left early.

Recommendation: Introduce a loyalty pricing model for high-paying customers. Retaining a customer paying \$73/month for 5 years generates \$4,380 in recurring revenue alone.

Query 5 — Top 10 Churn Reasons

Identifies the specific reasons customers gave for leaving, filtered to churned customers only.

```
SELECT
    churn_reason,
    churn_category,
    COUNT(customer_id) AS total_churned,
    ROUND(COUNT(customer_id) * 100.0 / SUM(COUNT(customer_id)) OVER(), 2) AS
percentage
FROM telecom_churn
WHERE customer_status = 'Churned'
GROUP BY churn_reason, churn_category
ORDER BY total_churned DESC
LIMIT 10;
```

Churn Reason	Category	Churned	%
Competitor had better devices	Competitor	313	16.75%
Competitor made better offer	Competitor	311	16.64%
Attitude of support person	Attitude	220	11.77%
Don't know	Other	130	6.96%
Competitor offered more data	Competitor	117	6.26%
Competitor offered higher speeds	Competitor	100	5.35%
Attitude of service provider	Attitude	94	5.03%
Price too high	Price	78	4.17%
Product dissatisfaction	Dissatisfaction	77	4.12%
Network reliability	Dissatisfaction	72	3.85%

Insight: Competitors account for 45% of all churn reasons

The top 4 competitor-related reasons alone account for 44.99% of all churn. Attitude-related reasons account for a further 16.8% — 314 customers left purely due to poor staff behaviour which is entirely preventable.

Recommendation: Deploy a competitive response strategy addressing device offers, data packages and download speeds. Implement urgent customer service training to address attitude-related churn.

Query 6 — Churn Rate by Payment Method

Examines whether payment method influences customer loyalty and churn behaviour.

```
SELECT
    payment_method,
```

```

COUNT(customer_id) AS total_customers,
SUM(CASE WHEN customer_status = 'Churned' THEN 1 ELSE 0 END) AS
churned_customers,
ROUND(SUM(CASE WHEN customer_status = 'Churned' THEN 1 ELSE 0 END) * 100.0 /
COUNT(customer_id), 2) AS churn_rate
FROM telecom_churn
GROUP BY payment_method
ORDER BY churn_rate DESC;

```

Payment Method	Total Customers	Churned	Churn Rate
Mailed Check	385	142	36.88%
Bank Withdrawal	3,909	1,329	34.00%
Credit Card	2,749	398	14.48%

Insight: Credit card customers churn at less than half the rate of others

Credit Card customers churn at just 14.48% compared to 36.88% for Mailed Check and 34.00% for Bank Withdrawal. Autopay creates passive commitment and reduces monthly billing friction.

Recommendation: Incentivize customers to switch to credit card autopay through a small monthly discount. This reduces churn friction and creates passive commitment.

Query 7 — Churn Rate by Offer

Evaluates which marketing offers are actually retaining customers vs accelerating churn.

```

SELECT
  offer,
  COUNT(customer_id) AS total_customers,
  SUM(CASE WHEN customer_status = 'Churned' THEN 1 ELSE 0 END) AS
churned_customers,
ROUND(SUM(CASE WHEN customer_status = 'Churned' THEN 1 ELSE 0 END) * 100.0 /
COUNT(customer_id), 2) AS churn_rate
FROM telecom_churn
GROUP BY offer
ORDER BY churn_rate DESC;

```

Offer	Total Customers	Churned	Churn Rate
Offer E	805	426	52.92%
None	3,877	1,051	27.11%
Offer D	602	161	26.74%
Offer C	415	95	22.89%
Offer B	824	101	12.26%
Offer A	520	35	6.73%

Insight: Offer E is driving churn — Offer A is the gold standard

Offer E has a 52.92% churn rate — higher than even Month-to-Month contracts. Meanwhile Offer A retains customers with just 6.73% churn. The gap between best and worst offer is enormous.

Recommendation: Immediately investigate and redesign Offer E. Scale Offer A and B across the customer base — especially targeting the 3,877 customers currently on no offer.

Query 8 — High Risk Customer Profile

Combines four dimensions simultaneously to identify the exact customer profiles with churn rates above 50%. Uses HAVING to filter aggregated results — an important distinction from WHERE which filters raw rows.

```
SELECT
    contract,
    internet_type,
    payment_method,
    customer_category,
    COUNT(customer_id) AS total_customers,
    SUM(CASE WHEN customer_status = 'Churned' THEN 1 ELSE 0 END) AS
    churned_customers,
    ROUND(SUM(CASE WHEN customer_status = 'Churned' THEN 1 ELSE 0 END) * 100.0 /
    COUNT(customer_id), 2) AS churn_rate
FROM telecom_churn
GROUP BY contract, internet_type, payment_method, customer_category
HAVING churn_rate >= 50
ORDER BY churn_rate DESC
LIMIT 10;
```

Contract	Internet	Payment	Category	Total	Churned	Churn Rate
Month-to-Month	Fiber Optic	Mailed Check	New Customer	42	35	83.33%
Month-to-Month	Fiber Optic	Bank Withdrawal	New Customer	655	496	75.73%
Month-to-Month	Cable	Mailed Check	New Customer	36	26	72.22%
Month-to-Month	Cable	Bank Withdrawal	New Customer	120	74	61.67%
Month-to-Month	DSL	Mailed Check	New Customer	43	26	60.47%
Month-to-Month	Fiber Optic	Credit Card	New Customer	142	84	59.15%
Month-to-Month	Fiber Optic	Bank Withdrawal	Developing	284	161	56.69%
Month-to-Month	Fiber Optic	Mailed Check	Developing	13	7	53.85%

Insight: Month-to-Month + Fiber Optic + New Customer = extreme churn risk

The single most at-risk segment is Month-to-Month + Fiber Optic + Bank Withdrawal + New Customer with 655 customers and a 75.73% churn rate — 496 customers already lost from this one segment alone.

Recommendation: Build a New Customer Retention Program targeting Fiber Optic Month-to-Month customers in their first 12 months. Offer incentives to upgrade to One Year contracts before month 6.

Query 9 — Churn Rate by Tenure Group

Analyses how customer loyalty and revenue evolve across different tenure groups. Combines two calculated columns created in Excel.

```

SELECT
    tenure_group,
    customer_category,
    COUNT(customer_id) AS total_customers,
    SUM(CASE WHEN customer_status = 'Churned' THEN 1 ELSE 0 END) AS
churned_customers,
    ROUND(SUM(CASE WHEN customer_status = 'Churned' THEN 1 ELSE 0 END) * 100.0 /
COUNT(customer_id), 2) AS churn_rate,
    ROUND(AVG(monthly_charge), 2) AS avg_monthly_charge,
    ROUND(AVG(total_revenue), 2) AS avg_total_revenue
FROM telecom_churn
GROUP BY tenure_group, customer_category
ORDER BY churn_rate DESC;

```

Tenure Group	Category	Total	Churned	Churn Rate	Avg Monthly	Avg Revenue
0-1 Year	New Customer	2,186	1,037	47.44%	\$54.95	\$387.91
1-2 Years	Developing Customer	1,024	294	28.71%	\$59.74	\$1,560.12
2-4 Years	Established Customer	1,594	325	20.39%	\$64.95	\$3,205.66
4-6 Years	Loyal Customer	2,239	213	9.51%	\$72.84	\$6,170.51

Insight: A Loyal Customer is worth 15.9x more than a New Customer

New customers churn at 47.44% while Loyal customers churn at just 9.51%. A Loyal Customer generates \$6,170.51 in total revenue vs \$387.91 for a New Customer — a 15.9x difference. Year one is the most critical retention period.

Recommendation: Calculate Customer Lifetime Value for year-one retention. Spending \$200-300 in retention incentives per new customer in year one is justified given the potential \$6,170 lifetime value.

Query 10 — Churn Rate by Age Group

Segments churn behaviour by age group to identify demographic patterns.

```

SELECT
    age_group,
    COUNT(customer_id) AS total_customers,
    SUM(CASE WHEN customer_status = 'Churned' THEN 1 ELSE 0 END) AS
churned_customers,
    ROUND(SUM(CASE WHEN customer_status = 'Churned' THEN 1 ELSE 0 END) * 100.0 /
COUNT(customer_id), 2) AS churn_rate,
    ROUND(AVG(monthly_charge), 2) AS avg_monthly_charge,
    ROUND(AVG(total_revenue), 2) AS avg_total_revenue
FROM telecom_churn
GROUP BY age_group
ORDER BY churn_rate DESC;

```

Age Group	Total	Churned	Churn Rate	Avg Monthly	Avg Revenue
61+	1,662	606	36.46%	\$72.56	\$3,338.21
46-60	1,909	464	24.31%	\$61.08	\$2,926.84
31-45	1,943	457	23.52%	\$60.07	\$2,922.10

18-30	1,529	342	22.37%	\$61.47	\$2,981.07
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Insight: Senior customers (61+) churn the most despite paying the most

The 61+ age group churns at 36.46% — 14 percentage points higher than the 18-30 group — while paying the highest average monthly charge of \$72.56. This mirrors the Fiber Optic paradox of premium pricing paired with high churn.

Recommendation: Create a Senior Customer Care Program with dedicated patient support, simplified plan options and proactive check-in calls for senior Month-to-Month customers.

Query 11 — Churn by Add-on Services

Uses UNION ALL to combine three separate service analyses into one result, showing whether subscribing to add-on services reduces churn.

```
SELECT 'Online Security' AS add_on_service, online_security AS subscribed,
       COUNT(customer_id) AS total_customers,
       SUM(CASE WHEN customer_status = 'Churned' THEN 1 ELSE 0 END) AS
churned_customers,
       ROUND(SUM(CASE WHEN customer_status = 'Churned' THEN 1 ELSE 0 END) * 100.0 /
COUNT(customer_id), 2) AS churn_rate
FROM telecom_churn GROUP BY online_security
UNION ALL
SELECT 'Online Backup' AS add_on_service, online_backup AS subscribed,
       COUNT(customer_id), SUM(CASE WHEN customer_status = 'Churned' THEN 1 ELSE 0
END),
       ROUND(SUM(CASE WHEN customer_status = 'Churned' THEN 1 ELSE 0 END) * 100.0 /
COUNT(customer_id), 2)
FROM telecom_churn GROUP BY online_backup
UNION ALL
SELECT 'Premium Tech Support' AS add_on_service, premium_tech_support AS
subscribed,
       COUNT(customer_id), SUM(CASE WHEN customer_status = 'Churned' THEN 1 ELSE 0
END),
       ROUND(SUM(CASE WHEN customer_status = 'Churned' THEN 1 ELSE 0 END) * 100.0 /
COUNT(customer_id), 2)
FROM telecom_churn GROUP BY premium_tech_support
ORDER BY add_on_service, churn_rate DESC;
```

Add-on Service	Subscribed	Total	Churned	Churn Rate
Online Security	No	3,498	1,461	41.77%
Online Security	Yes	2,019	295	14.61%
Premium Tech Support	No	3,473	1,446	41.64%
Premium Tech Support	Yes	2,044	310	15.17%
Online Backup	No	3,088	1,233	39.93%
Online Backup	Yes	2,429	523	21.53%

Insight: Add-on services reduce churn by up to 65%

Online Security reduces churn from 41.77% to 14.61% — a 65% reduction. Premium Tech Support reduces churn from 41.64% to 15.17% — a 64% reduction. Customers with add-ons have more perceived value and higher switching costs.

Recommendation: Bundle add-on services into new customer onboarding packages. Offer Online Security free for the first 3 months to Fiber Optic Month-to-Month customers — the highest risk segment.

Query 12 — Churn by Number of Referrals

Analyses whether customers who refer others are more loyal — using referral count as a proxy for satisfaction.

```
SELECT
    number_of_referrals,
    COUNT(customer_id) AS total_customers,
    SUM(CASE WHEN customer_status = 'Churned' THEN 1 ELSE 0 END) AS
churned_customers,
    ROUND(SUM(CASE WHEN customer_status = 'Churned' THEN 1 ELSE 0 END) * 100.0 /
COUNT(customer_id), 2) AS churn_rate,
    ROUND(AVG(total_revenue), 2) AS avg_total_revenue
FROM telecom_churn
GROUP BY number_of_referrals
ORDER BY number_of_referrals ASC;
```

Referrals	Total Customers	Churned	Churn Rate	Avg Revenue
0	3,821	1,245	32.58%	\$2,206.24
1	1,086	507	46.69%	\$3,618.71
2	2,362	26	11.02%	\$4,227.51
4	236	18	7.63%	\$4,494.28
6	221	83	3.62%	\$4,006.98
8	213	20	0.94%	\$4,043.83
10	223	0	0.00%	\$4,271.26

Insight: 10 referrals = 0% churn — referrers are 2x more valuable

Churn drops consistently as referrals increase — customers with 10 referrals have 0% churn. Referring customers also generate 2x more revenue (\$4,271 vs \$2,206 for 0 referrals).

Recommendation: Build a structured referral rewards program. Incentivizing referrals doesn't just bring new customers — it locks in the referring customer making them dramatically less likely to churn.

Query 13 — Top 15 Cities by Revenue Lost to Churn

Identifies geographic hotspots of churn using a revenue-weighted CASE WHEN — summing actual revenue values instead of counting rows. Uses HAVING to filter cities with meaningful sample sizes.

```
SELECT
    city,
    COUNT(customer_id) AS total_customers,
    SUM(CASE WHEN customer_status = 'Churned' THEN 1 ELSE 0 END) AS
churned_customers,
    ROUND(SUM(CASE WHEN customer_status = 'Churned' THEN 1 ELSE 0 END) * 100.0 /
COUNT(customer_id), 2) AS churn_rate,
```

```
ROUND(SUM(CASE WHEN customer_status = 'Churned' THEN total_revenue ELSE 0
END), 2) AS revenue_lost,
ROUND(AVG(CASE WHEN customer_status = 'Churned' THEN monthly_charge ELSE NULL
END), 2) AS avg_monthly_charge_churned
FROM telecom_churn
GROUP BY city
HAVING churned_customers >= 10
ORDER BY revenue_lost DESC
LIMIT 15;
```

City	Total	Churned	Churn Rate	Revenue Lost	Avg Monthly
San Diego	285	185	64.91%	\$385,446.39	\$72.97
Los Angeles	293	78	26.62%	\$147,090.46	\$73.54
Sacramento	108	26	24.07%	\$59,449.20	\$72.74
San Francisco	104	31	29.81%	\$56,894.27	\$79.89
Santa Barbara	28	10	35.71%	\$46,033.71	\$86.07
San Jose	112	29	25.89%	\$44,393.49	\$70.05
Temecula	38	22	57.89%	\$44,102.41	\$81.18
Escondido	51	16	31.37%	\$38,869.89	\$80.09
Fallbrook	43	26	60.47%	\$34,691.78	\$70.21
Fresno	61	13	21.31%	\$29,607.42	\$67.97

Insight: San Diego is a crisis market — \$385K revenue lost

San Diego has the highest revenue loss at \$385,446.39 with a catastrophic 64.91% churn rate — nearly 2 in 3 San Diego customers are leaving. Temecula (57.89%) and Fallbrook (60.47%) also show alarming rates suggesting localized competitive pressure.

Recommendation: Deploy city-specific retention campaigns starting with San Diego. Investigate whether a specific competitor has entered these markets aggressively and respond with targeted counter-offers.

4. Stage 2 Summary — Key Findings

#	Analysis	Key Finding
1	Overall churn rate	26.54% — above industry average
2	By contract type	Month-to-Month: 45.84% churn (18x Two Year)
3	By internet type	Fiber Optic: 40.72% churn
4	Revenue lost	\$3.68M total · \$137K/month recurring
5	Top churn reasons	Competitors: 45% · Attitude: 16.8%
6	By payment method	Credit Card: 14.48% vs Mailed Check: 36.88%
7	By offer	Offer E: 52.92% · Offer A: 6.73%
8	High risk profile	M-t-M + Fiber Optic + New Customer = 75%+

9	By tenure group	New customers: 47.44% · Loyal: 9.51%
10	By age group	61+ customers: 36.46% churn
11	By add-on services	Online Security reduces churn by 65%
12	By referrals	10 referrals = 0% churn · 2x more revenue
13	By city	San Diego: \$385K lost · 64.91% churn rate

Next Stage

Stage 3 — Power BI: Build a 3-page interactive dashboard telling the complete churn story. Page 1: What is happening? | Page 2: Why is it happening? | Page 3: Who is at risk and what does it cost?